

Lab 03

Agile / Scrum – Jira

Project: Webhook Ingestion & Retry Mechanism Hub

Name	Prathyush Gowda
SRN	PES1UG24AM201
Agile Tool	Jira

This report documents the Agile/Scrum implementation of the Webhook Ingestion & Retry Mechanism Hub in Jira, including the product backlog, user stories, story-point estimation, sprint planning, sprint execution, burndown progress, and overall project status.

1. Epics and User Stories

Epic 1: Webhook Ingestion & Validation

Description: Enable reliable receipt, validation, and recording of inbound webhook events before they enter the processing pipeline.

User Story	As a / I want to / So that	Priority	Points
Receive Inbound Webhooks	As an integration developer, I want the system to receive inbound webhooks so that third-party events can enter the processing pipeline.	Highest	5
Validate Webhook Payloads	As an integration developer, I want the system to validate incoming webhook payloads so that invalid requests are rejected before processing.	Highest	5

Epic 2: Webhook Delivery

Description: Enable valid webhook payloads to be forwarded to configured target internal services and record delivery responses.

User Story	As a / I want to / So that	Priority	Points
Forward Valid Webhook Payloads	As an integration developer, I want valid webhook payloads forwarded to the configured target service so that third-party events reach the required internal service.	Highest	5
Record Delivery Responses	As an integration developer, I want delivery responses recorded so that successful and failed webhook deliveries can be tracked and audited.	High	3

Epic 3: Retry & Reliability

Description: Provide automatic retry handling, exponential backoff, and preservation of webhook processing and retry state.

User Story	As a / I want to / So that	Priority	Points
Retry Failed Webhook Deliveries	As an integration developer, I want failed webhook deliveries retried automatically so that temporary service failures do not cause event loss.	Highest	5
Implement Exponential Backoff	As an integration developer, I want failed webhook retries to use increasing exponential backoff intervals so that repeated failures do not overload the target service.	High	5

1. Epics and User Stories (continued)

Epic 4: Logging & Monitoring

Description: Maintain detailed webhook, delivery attempt, response, and processing status logs for monitoring and troubleshooting.

User Story	As a / I want to / So that	Priority	Points
Log Webhook Processing Details	As a system operator, I want webhook payload details and processing status recorded in logs so that I can monitor and troubleshoot webhook events.	High	3
Track Delivery Attempts	As a system operator, I want delivery attempts and responses recorded so that I can monitor delivery behavior and investigate failures.	High	3

Epic 5: Dead-Letter Queue Management

Description: Store repeatedly failed webhook events in a dead-letter queue for review and controlled reprocessing.

User Story	As a / I want to / So that	Priority	Points
Move Failed Events to Dead-Letter Queue	As a system operator, I want webhook events that exceed the retry limit moved to a dead-letter queue so that failed events are not permanently lost.	High	5
Review and Reprocess Dead-Letter Events	As a system operator, I want dead-letter events available for review and controlled reprocessing so that unresolved webhook events can be handled later.	Medium	5

2. Story Points and Planning Poker Estimation

Story points were assigned using a Fibonacci-style Planning Poker approach. The ten stories use 3- and 5-point estimates based on relative complexity, implementation effort, and uncertainty.

Points	Stories	Count	Total
3	Record Delivery Responses; Log Webhook Processing Details; Track Delivery Attempts	3	9
5	Receive Inbound Webhooks; Validate Webhook Payloads; Forward Valid Webhook Payloads; Retry Failed Webhook Deliveries; Implement Exponential Backoff; Move Failed Events to Dead-Letter Queue; Review and Reprocess Dead-Letter Events	7	35
Total	10 User Stories	10	44

3. Priority and Product Backlog

The Jira backlog contains five Epics and ten User Stories. The stories are prioritized as Highest, High, and Medium according to their importance to the webhook processing workflow.

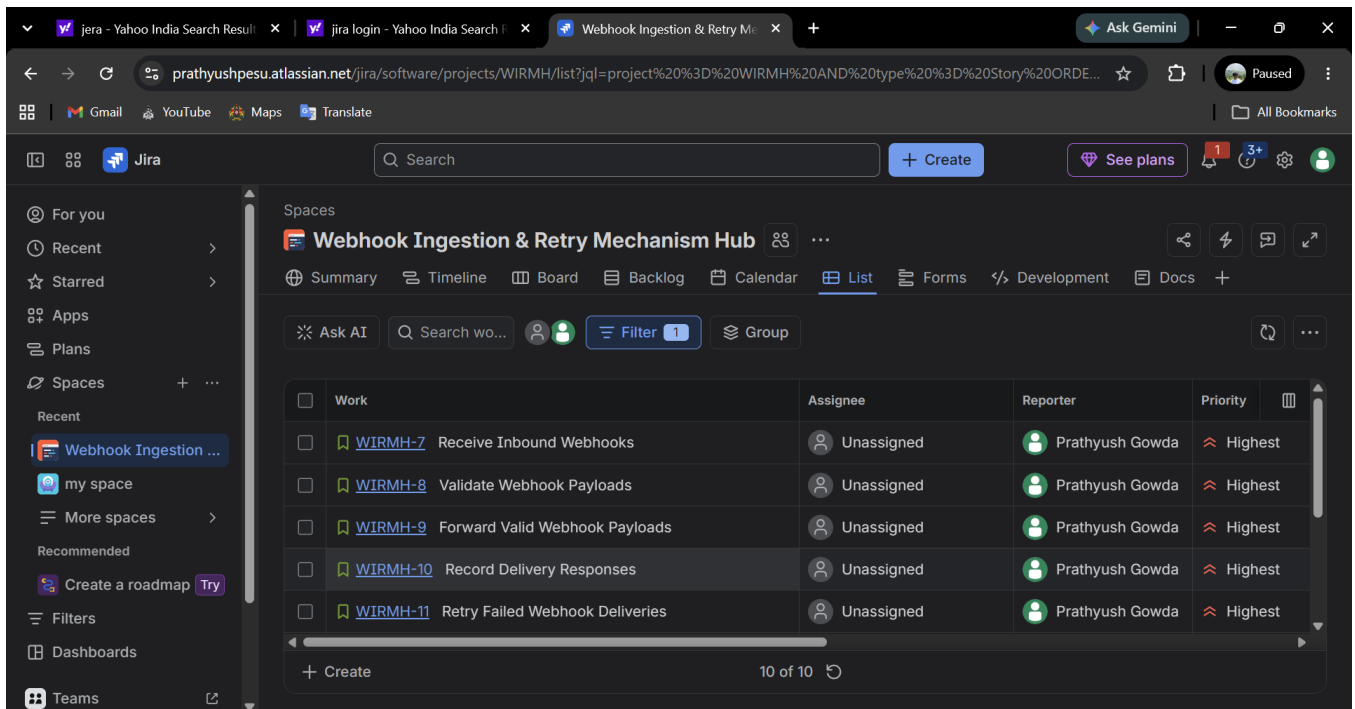


Figure 1: Jira List view showing the webhook user stories with priority and status fields.

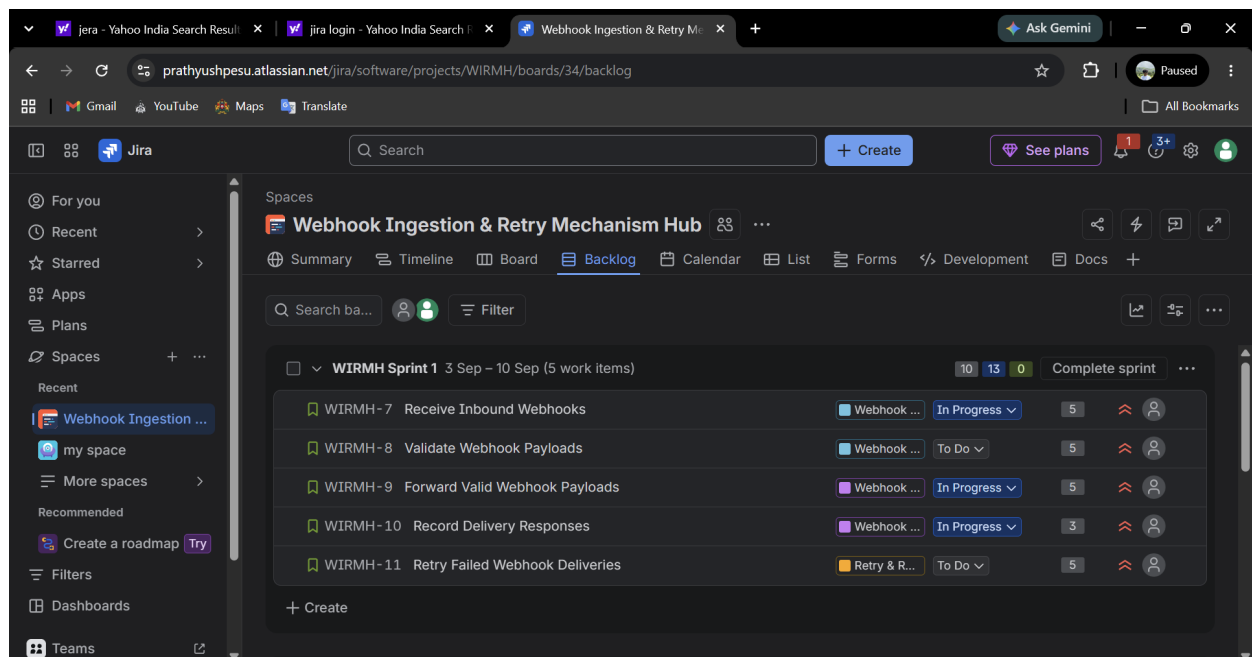


Figure 2: Jira view showing the five Epics created for the Webhook Ingestion & Retry Mechanism Hub.

4. Sprint Planning and Execution

The ten stories were divided across two one-week Sprints. Sprint 1 contains the core ingestion and delivery work, while Sprint 2 contains reliability, monitoring, and dead-letter queue work. The Jira screenshots show the sprint dates as 3 Sep – 10 Sep.

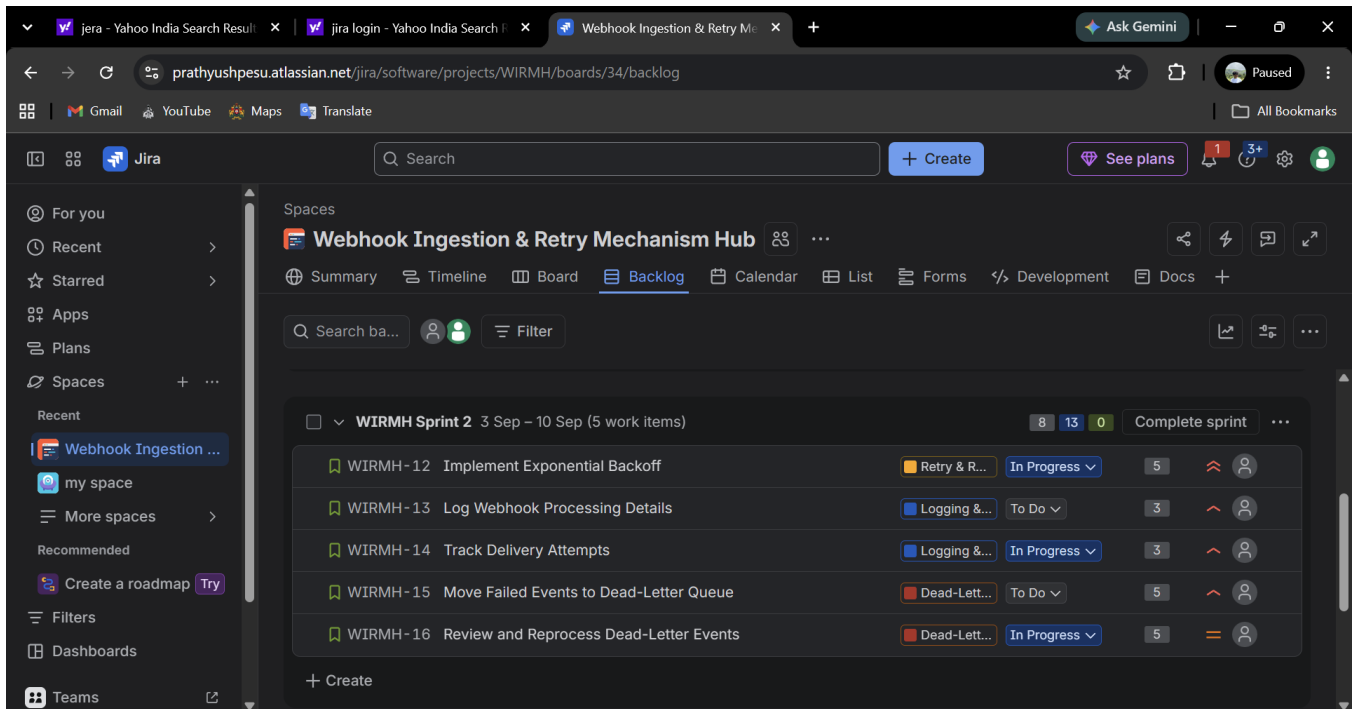


Figure 3: Sprint 2 planned in the Jira backlog with five work items.

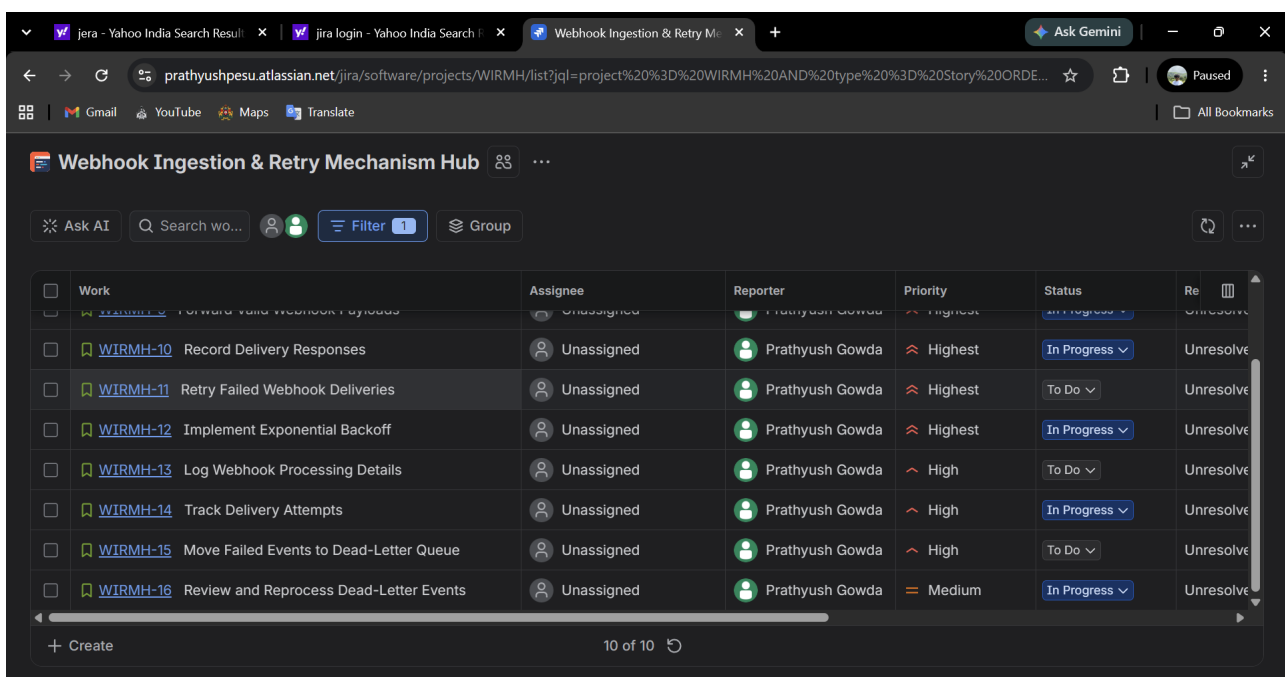


Figure 4: Sprint 1 planned in the Jira backlog with five work items and 23 story points.

Sprint allocation: Sprint 1 = 5 stories / 23 points. Sprint 2 = 5 stories / 21 points.

5. Sprint 1

Sprint 1 – Core Webhook Processing was executed with five stories. The Jira board was used to track work through the Scrum workflow.

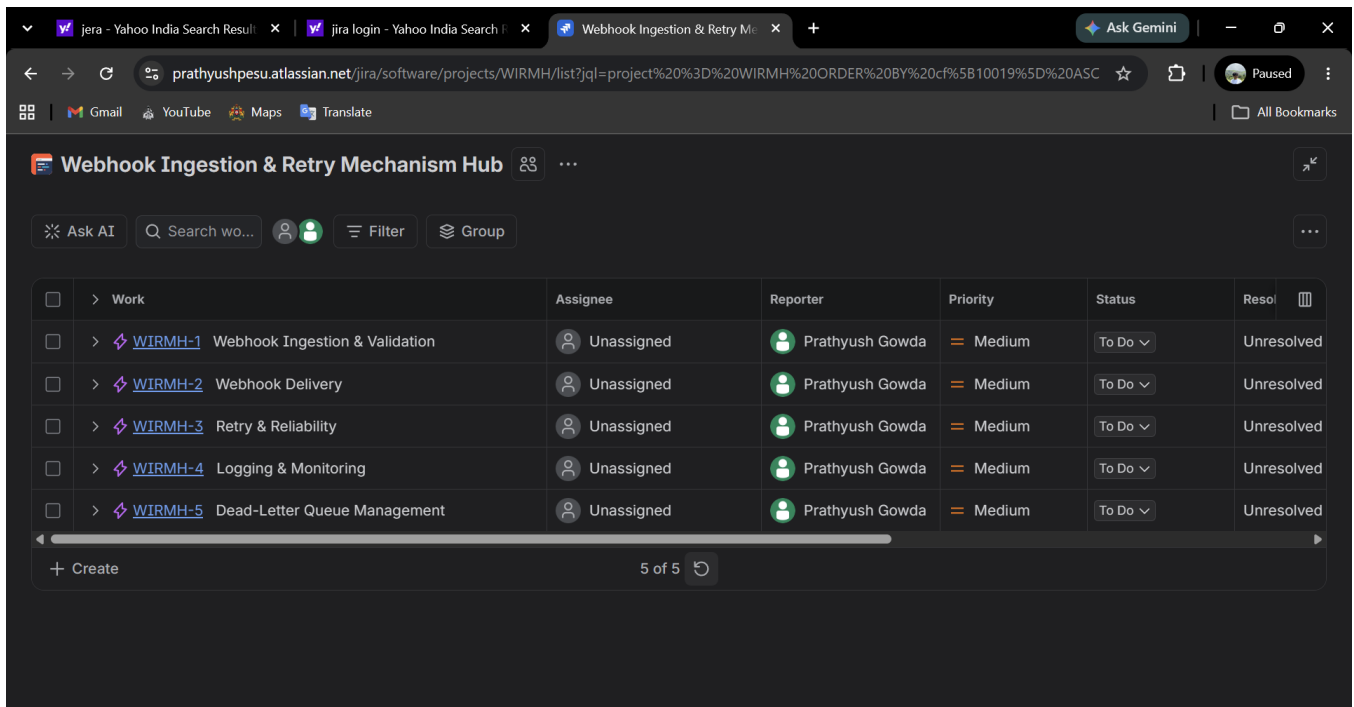


Figure 5: Jira sprint board/list evidence for Sprint 1 work items.

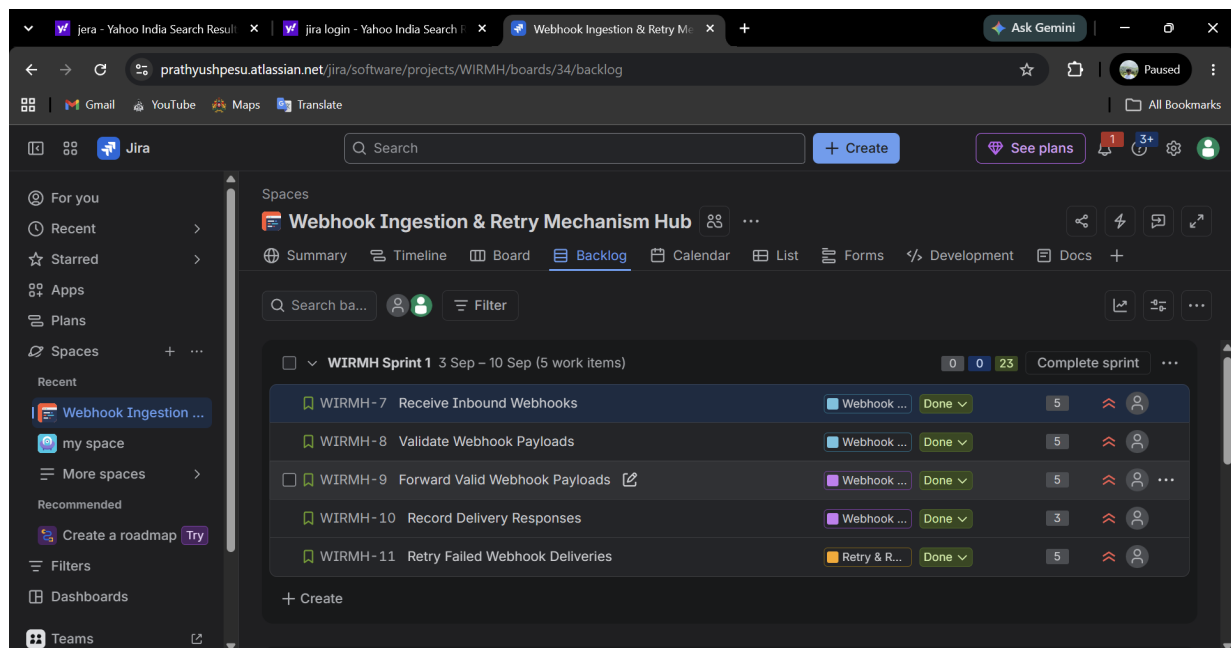


Figure 6: Jira progress evidence associated with the sprint execution.

6. Sprint 2

Sprint 2 – Reliability, Monitoring & Dead-Letter Handling contains the remaining five stories covering exponential backoff, logging, delivery-attempt tracking, dead-letter handling, and reprocessing.

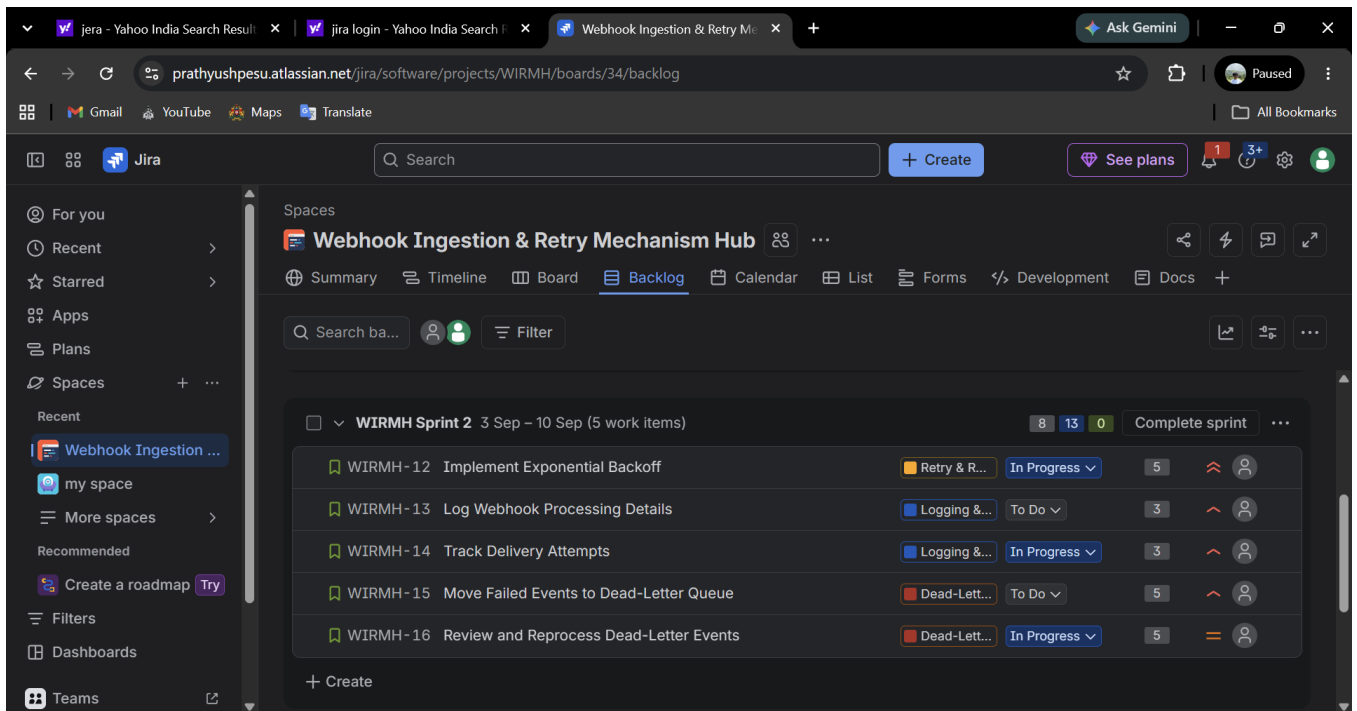


Figure 7: Sprint 2 in the Jira backlog with five work items and 21 story points.

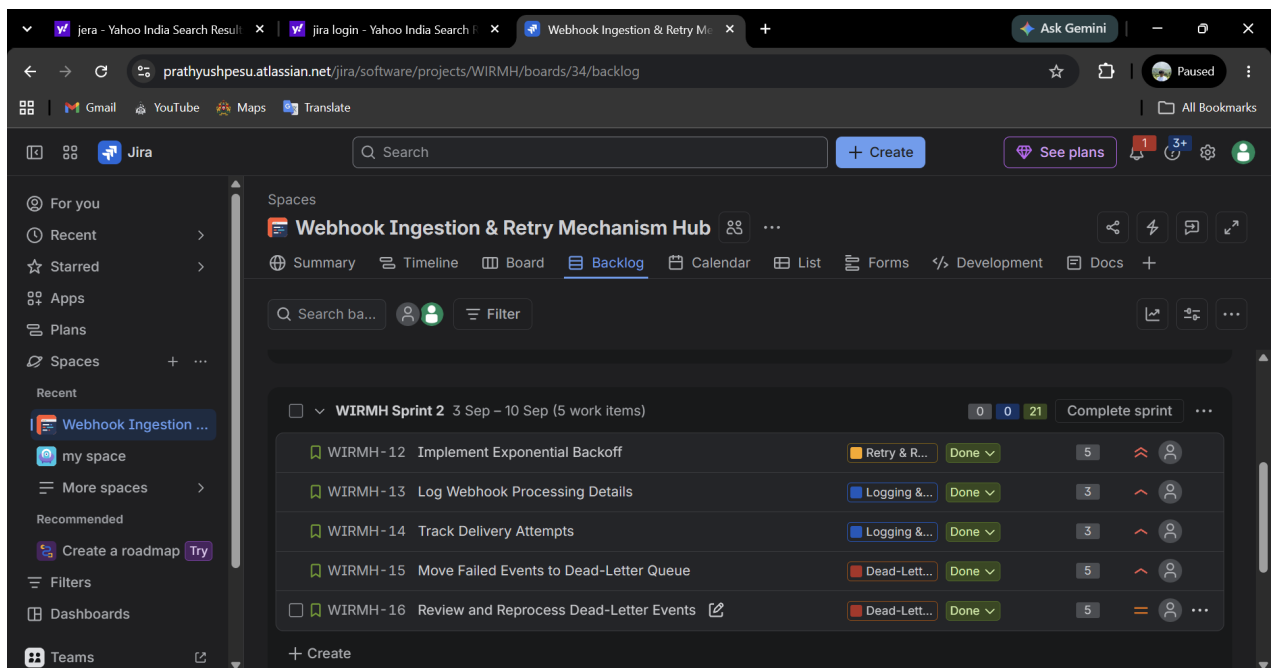
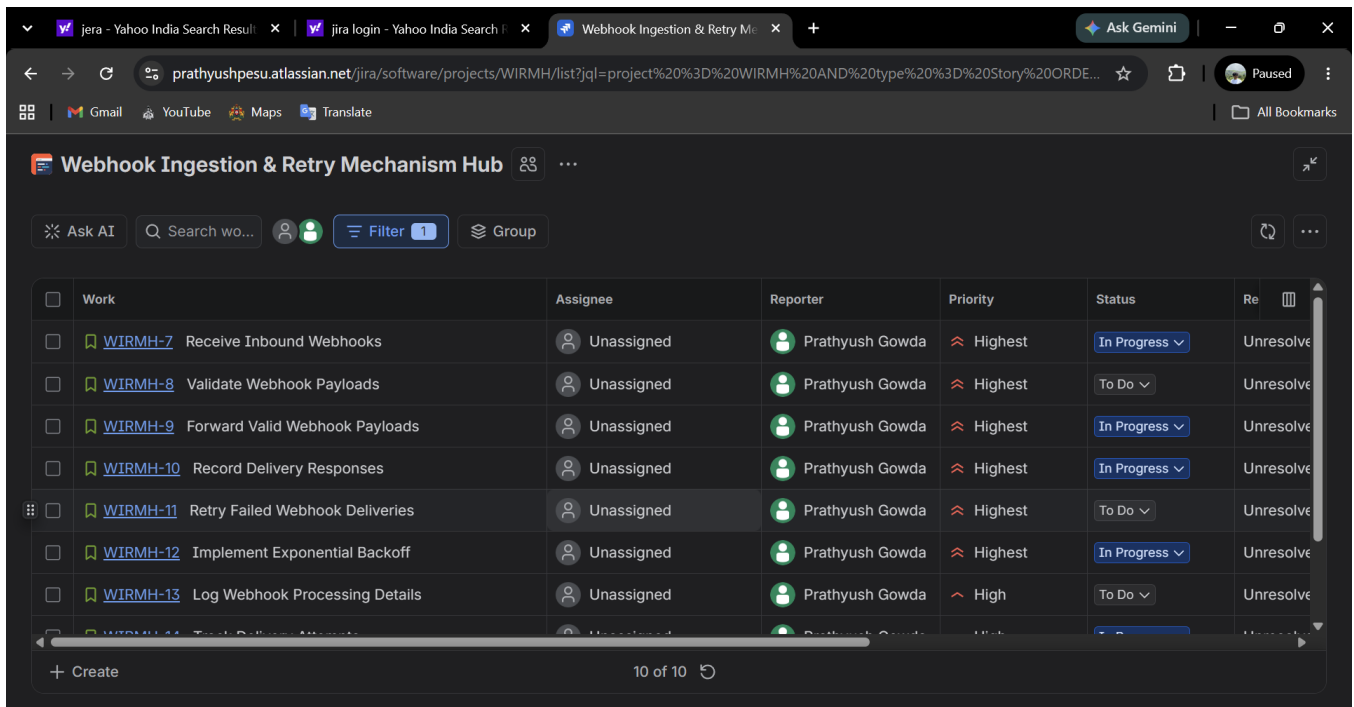


Figure 8: Jira sprint board showing the Sprint 2 stories in workflow.

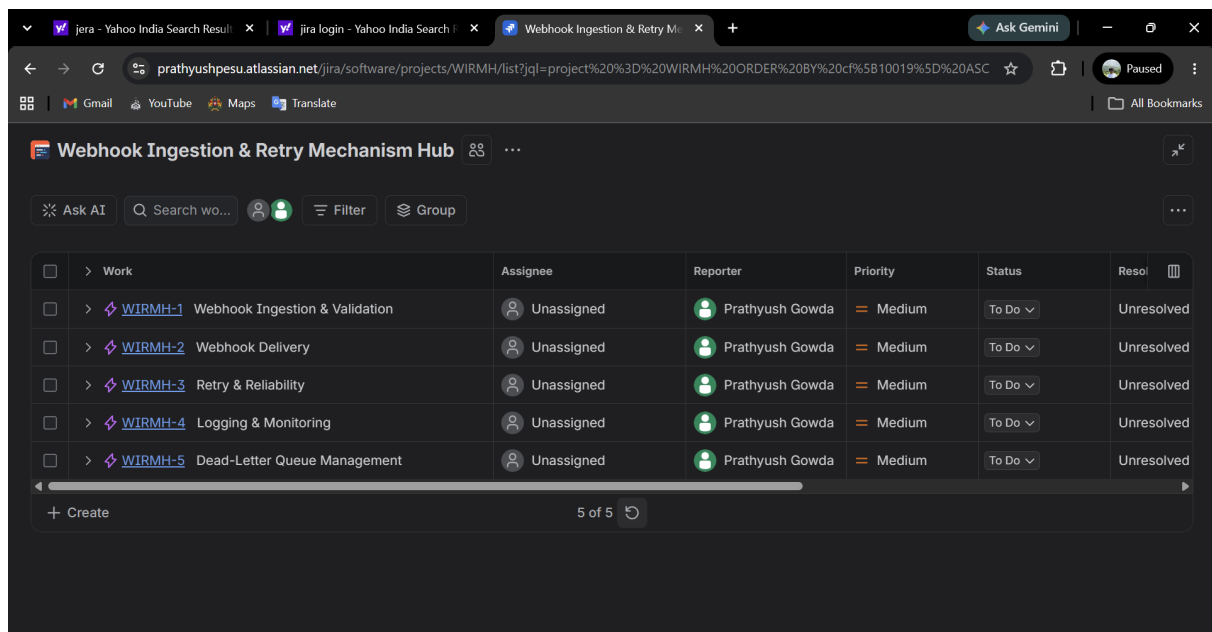
7. Scrum Workflow Progress

The Jira evidence shows work items progressing through the standard Scrum states To Do, In Progress, and Done. This demonstrates the execution and tracking of sprint work.



Work	Assignee	Reporter	Priority	Status	Resolution
WIRMH-7 Receive Inbound Webhooks	Unassigned	Prathyush Gowda	Highest	In Progress	Unresolved
WIRMH-8 Validate Webhook Payloads	Unassigned	Prathyush Gowda	Highest	To Do	Unresolved
WIRMH-9 Forward Valid Webhook Payloads	Unassigned	Prathyush Gowda	Highest	In Progress	Unresolved
WIRMH-10 Record Delivery Responses	Unassigned	Prathyush Gowda	Highest	In Progress	Unresolved
WIRMH-11 Retry Failed Webhook Deliveries	Unassigned	Prathyush Gowda	Highest	To Do	Unresolved
WIRMH-12 Implement Exponential Backoff	Unassigned	Prathyush Gowda	Highest	In Progress	Unresolved
WIRMH-13 Log Webhook Processing Details	Unassigned	Prathyush Gowda	High	To Do	Unresolved

Figure 9: Jira List view showing user stories with workflow status values.



Work	Assignee	Reporter	Priority	Status	Resolution
WIRMH-1 Webhook Ingestion & Validation	Unassigned	Prathyush Gowda	Medium	To Do	Unresolved
WIRMH-2 Webhook Delivery	Unassigned	Prathyush Gowda	Medium	To Do	Unresolved
WIRMH-3 Retry & Reliability	Unassigned	Prathyush Gowda	Medium	To Do	Unresolved
WIRMH-4 Logging & Monitoring	Unassigned	Prathyush Gowda	Medium	To Do	Unresolved
WIRMH-5 Dead-Letter Queue Management	Unassigned	Prathyush Gowda	Medium	To Do	Unresolved

Figure 10: Jira backlog/board evidence showing completed and in-progress work items.

8. Comments / Sprint Updates

Jira comments can be used to communicate implementation updates and keep the work-item history clear. A suitable update for this project is:

Suggested Jira comment
Implemented webhook processing flow with validation, delivery handling, retry support, and status tracking aligned with the sprint requirements.

The supplied Jira screenshots do not show a dedicated comment-entry screen, so this section documents the appropriate sprint-update text rather than claiming a screenshot of a posted comment.

9. Burndown Chart / Sprint Progress

The Jira burndown report compares remaining story points with the ideal guideline across the sprint dates and provides a visual indication of sprint progress.

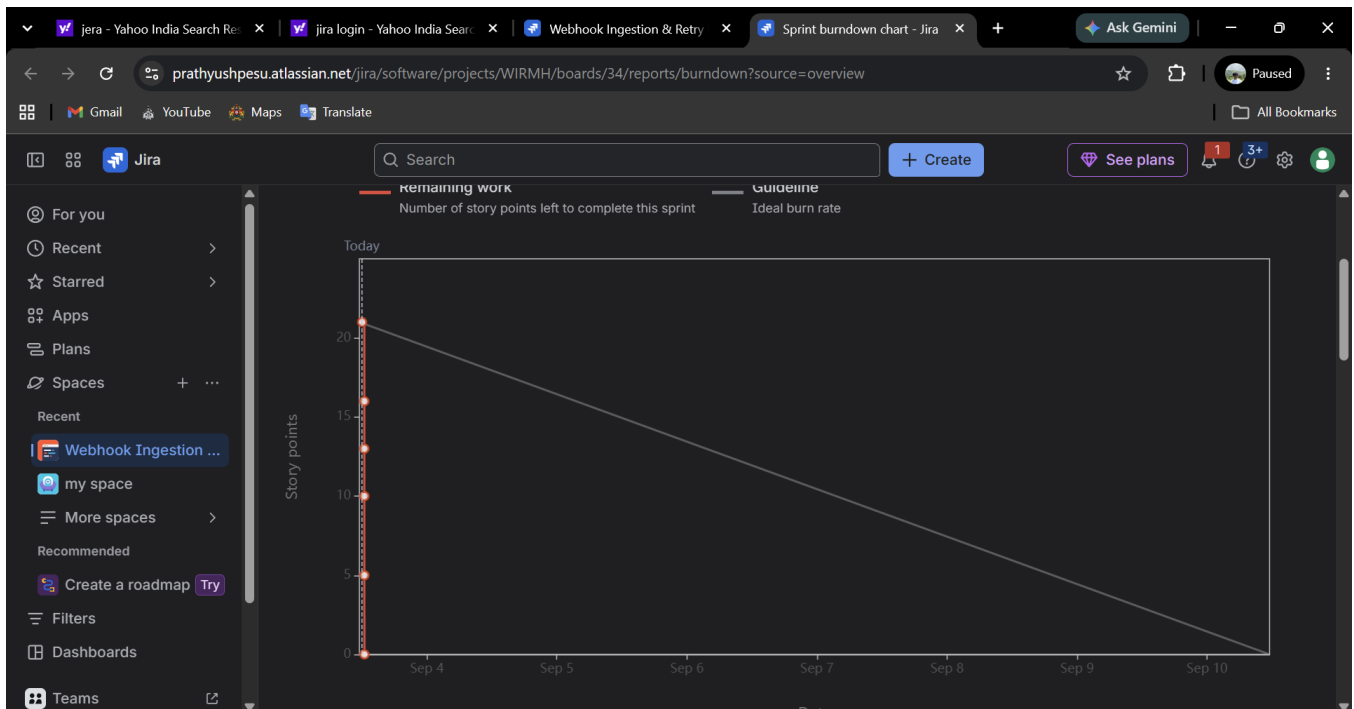


Figure 11: Jira Sprint Burndown Chart showing remaining work and the ideal guideline.

10. Summary / Overall Jira View

The Jira Summary dashboard provides an overall view of project status, priority distribution, work-item types, and Epic progress.



Figure 12: Jira Summary dashboard for the Webhook Ingestion & Retry Mechanism Hub.

11. Final Reflection

This lab provided practical experience with Agile and Scrum using Jira for the Webhook Ingestion & Retry Mechanism Hub. I learned how to break a software requirement into Epics and User Stories, define clear user-focused requirements, assign priorities and story points, and organize the backlog into time-bounded Sprints.

Planning the work across two Sprints helped me understand how a Scrum team can deliver functionality incrementally. Sprint 1 focused on the core webhook ingestion and delivery flow, while Sprint 2 focused on reliability, monitoring, and dead-letter processing. Moving work items through To Do, In Progress, and Done made the Scrum workflow more practical.

The burndown chart provided a useful view of remaining work against the ideal guideline, while the Jira Summary dashboard gave an overall project-level picture of the backlog and Epic progress. Overall, the exercise made Agile concepts such as backlog prioritization, estimation, sprint planning, execution, and progress tracking easier to understand.

Project: Webhook Ingestion & Retry Mechanism Hub

Name: Prathyush Gowda

SRN: PES1UG24AM201